Knowledge-Aided Momentum Contrastive Learning for Remote-Sensing Image Text Retrieval

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***Abstract*— Remote-sensing image–text retrieval (RSITR) has attracted widespread attention due to its great potential for rapid information mining ability on remote-sensing images. Although significant progress has been achieved, existing methods typically overlook the challenge posed by the extremely analogous descriptions, where the subtle differences remain largely unexploited or, in some cases, are entirely disregarded. To address the limitation, we propose a knowledge-aided momentum contrastive learning (KAMCL) method for RSITR. Specifically, we propose a novel knowledge-aided learning (KAL) framework, including knowledge initialization, construction, filtration, and alignment operations, which aims at providing valuable concepts and learning discriminative representations. On this basis, we integrate momentum contrastive learning (MCL) to promote the capture of key concepts within the representation via expanding the scale of negative sample pairs. Moreover, we designed a hierarchical aggregator (HA) module to better capture the multilevel information from remote-sensing images. Finally, we introduce an innovative two-step training strategy designed to effectively harness the synergy among concepts and leverage their respective functionalities. Extensive experiments conducted on the three public datasets showcase the remarkable performance of our approach in terms of retrieval accuracy and computational efficiency. For instance, compared with the existing state-of-the-art method, our method exhibits notable performance improvements of 2.65% on the RSICD dataset, simultaneously achieving improvements in inference efficiency by 48%. Our source code will be released at https://github.com/mcx-mcx/KAMCL.**

***Index Terms*— Hierarchical aggregator (HA), knowledge-aided learning (KAL), momentum contrastive learning (MCL), remote-sensing image–text retrieval (RSITR).**

I. Introduction

ADVANCED remote-sensing technologies have been employed in disaster monitoring, military reconnaissance, geolocalization, and navigation with their vigorous developments [1], [2], [3]. In these applications, remote-sensing image–text retrieval (RSITR) plays a crucial role, which aims at searching for the remote-sensing images with similar semantic information given a text as query, and vice versa [4]. In response, researchers have turned their focus toward RSITR, recognizing its importance within the realm of remote-sensing multimodality [5], [6].

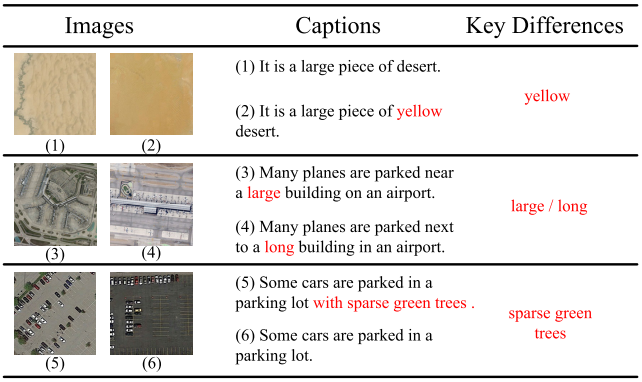


Fig. 1. Three groups of similar remote-sensing image–text pairs, where the images and captions possess extremely similar semantics and cues except for a few key differences. Take the first group as an example, both the two images depict "desert," while the key difference is the adjective "yellow." However, the traditional methods [4], [17] fail to learn precise representations owing to a lack of guidance of key information.

Despite the remarkable advancements that have been achieved in various retrieval tasks, such as image–text retrieval [7], [8], [9], text-based person search [10], [11], and product search [12], there are distinct gaps between these domains and remote-sensing domain. This is attributed to the unique characteristics of remote-sensing image styles and text descriptions, such as a larger intraclass variance and a smaller interclass variance [13], [14], and the indistinguishable text descriptions. To bridge the semantic gap between remote-sensing images and text, researchers have made significant strides [4], [15], [16] regarding the aforementioned issues.

Although considerable efforts have been made to mitigate the impact of irrelevant objects [18], [19] and resolution disparities [20] in remote-sensing images, some challenges remain unexplored: the analogous descriptions in text modality pose obstacles to learning discriminative representations due to the intrinsic properties of the remote-sensing domain. It means that for the hard negative sample pairs, the corresponding text descriptions may be semantically similar with only the

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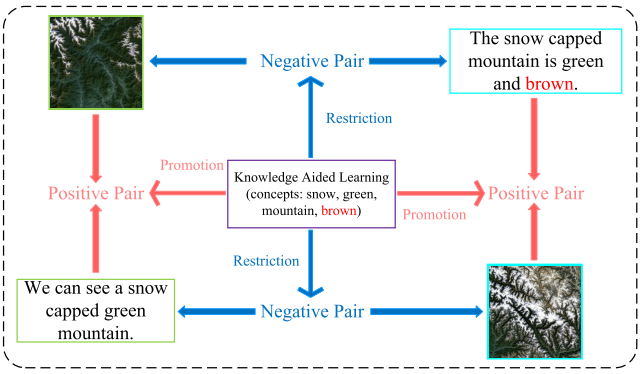


Fig. 2. Intuitive illustration of our proposed KAL framework. Traditional methods [4], [17] only obtain latent representations by calculating the similarity of positive and negative samples, while KAL aims at achieving more distinctive representations by emphasizing key concepts, which further distinguish the key differences between the positive pair and the negative pair. For example, the above two groups of image–text pairs are hard negative samples to each other, where our proposed KAL promotes closing the gap between the positive pair and restricting the gap between the negative pair with the help of the key concepts, for example, "brown."

difference of one word or phrase. As shown in Fig. 1, the text annotations for three groups of remote-sensing images are extremely similar. What is more, the key differences between the positive sample pair and the negative sample pair may consist of only one-word variation. This difficulty in description arises from the fact that remote-sensing images inherently possess a limited set of attributes, often resulting in a scarcity of words that effectively describe their features.

To overcome this challenge, we approach the issue from two perspectives. First, we enhance the differentiation of these confusing samples by further emphasizing these key differences. Existing methods [4], [17] primarily focus on obtaining distinctive cross-modal representations solely within the latent space, overlooking the reinforcement of key concepts at the level of representation. This leads to the absence of clear differentiation for these key concepts, which further exacerbates the disregard for the key differences. Second, the key to addressing this issue lies in large-scale contrastive learning. As shown in Fig. 1, apart from text, remote-sensing images also exhibit high similarity among themselves. For instance, distinguishing images of "deserts" in the first group could be challenging, due to their similar morphologies. By simultaneously learning from these hard samples, the model will be able to obtain more discriminative representations. The mini-batch-based training paradigm is suboptimal, inherently flawed, and imposes constraints on the acquisition of learnable negative sample pairs [21].

To this end, we propose a knowledge-aided momentum contrastive learning (KAMCL) method for RSITR. Specifically, we develop a novel knowledge-aided learning (KAL) framework, including knowledge initialization, construction, filtration, and alignment operations. As illustrated in Fig. 2, KAL aims at directly discerning indispensable key concepts at the level of representation for promoting positive pairs and restricting negative pairs, whereas the conventional approaches [4], [17] learn less discriminative representations since they fail to highlight the key differences in the remote-sensing domain. On this basis, we further design a two-step strategy for effectively obtaining discriminative representations from the momentum contrastive learning (MCL) and maximizing the effect of KAL. Additionally, we introduce a hierarchical aggregator (HA) to capture multilevel and comprehensive visual representations. We validate the effectiveness of our proposed KAMCL method on three well-established benchmarks: the RSICD dataset [15], RSITMD dataset [17], and NWPU-Captions dataset [22]. Extensive experiments demonstrate that KAMCL remarkably outperforms the state-of-the-art approaches.

The main contributions of our work are as follows.

1) We propose a KAMCL method for RSITR, which addresses the challenge of analogous descriptions in the remote-sensing domain from two perspectives: a novel KAL framework for reinforcing key concepts, and a two-step training strategy for effectively obtaining discriminative representations from MCL and leveraging the knowledge.
2) We develop a KAL framework, including knowledge initialization, construction, filtration, and alignment operations, to achieve a better focus on the key concepts. By enhancing the discriminative ability of key differences, the KAL framework further differentiates hard negative sample pairs.
3) We design an HA mechanism for capturing multilevel and comprehensive visual representations. This mechanism effectively integrates both local and global information from remote-sensing images, leading to more comprehensive visual representations.
4) Extensive experiments on three public benchmark datasets, that is, RSICD [15], RSITMD [17], and NWPU-Captions [22], show that KAMCL consistently outperforms the state-of-the-arts by a large margin.

The remaining sections of this article are organized as follows. In Section II, we provide an overview of related work on RSITR, contrastive learning, and KAL. The structure of our model and the description of the approaches are explained in detail in Section III. Subsequently, Section IV presents a comprehensive comparison between our proposed model and other methods, accompanied by extensive ablation experiments that investigate the role of each module. Finally, in Section V, we draw conclusions based on our findings.

II. Related Work

A. Remote-Sensing Image–Text Retrieval

Remote-sensing image–text retrieval typically adopts a dual-encoder architecture, where different modalities are encoded separately into a common embedding space for similarity calculation. For instance, Abdullah et al. [23] proposed a deep bidirectional triplet network with an average fusion strategy to combine features in the embedding space, which enables learning more robust representations. Yuan et al. [17] came up with a multiscale visual self-attention module to cope with the challenges of multiscale scarcity and target redundancy. Cheng et al. [24] introduced a deep semantic alignment module, which employs the attention and gate mechanisms

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to filter and optimize data features for further obtaining more discriminative representations. Lv et al. [25] proposed fusion-based correlation learning to capture the intermodality complementary information and aggregate the features accordingly. Subsequently, researchers direct their attention toward the utilization and aggregation of local information. Yuan et al. [26] designed a hidden supervised optimization method based on knowledge distillation, which incorporates the multiscale information and dynamically filters out redundant features to improve the model's representation. Following [17], Yuan et al. [4] introduced a multilevel information dynamic fusion module to aggregate the global and local features by a graph convolution network. Zheng et al. [27] alleviated redundant noise of local information and merged the disentangled scale and semantics in a union model. Mi et al. [28] integrated relevant information from the external knowledge graph to address the information asymmetry between images and text. To reduce semantic confusion, Pan et al. [16] amalgamated visual features across various scales and different granularity, thereby attaining scene-aware visual aggregation representations. Differing from others, we propose a novel KAMCL method that delves deeper into addressing previously overlooked the problem of analogous descriptions within the text modality.

B. Contrastive Learning

Contrastive learning aims at maximizing the similarities between related samples while minimizing those between unrelated samples, which has been widely applied in unsupervised visual representation learning [29], [30] and vision-language pretraining [31], [32], and so on. For instance, Chen et al. [33] designed a contrastive guidance branch to constrain the embeddings of different exemplars to enhance the process of rain removal. In contrast, Xiao et al. [34] integrated contrastive learning, treating the features from different encoders as positive pairs, as a guiding mechanism in the reconstruction process for blind remote-sensing image super-resolution. To overcome the limitations of traditional end-to-end training paradigms, Wu et al. [30] proposed the mechanism of memory bank, allowing for an extensive comparison of negative samples and thereby eliminating concerns regarding limited graphic processing unit (GPU) resources. He et al. [29] proposed a novel momentum contrastive (MoCo) training paradigm, which incorporates a momentum-updated key encoder and a dynamic queue to obtain substantial and consistent representations on the fly. Although SimCLR [35] achieves commendable performance without memory banks by utilizing larger batch sizes, appropriate data augmentations, and nonlinear transformation, MoCo v2 [36] demonstrates that the integration of these techniques with memory bank yields superior performance. Since then, many works have attempted to incorporate MoCo into both image and text modalities for their downstream tasks. For instance, Liu et al. [37] employed MoCo with four memory banks for text and images at both the feature-level and semantic-level matching and applied the InfoNCE loss within a hierarchical transformer structure, contributing to the attainment of more fine-grained representations for video–text retrieval. Wang et al. [38] extended MoCo to a diversity-sensitive contrastive learning paradigm for image–text retrieval, which adaptively weights image–text pairs by their semantic differences. Unlike previous methods that solely employ momentum contrast learning in their separate spaces and rely on extra memory banks to store representations of different levels, we extend MoCo to the hybrid space. This integration brings both semantic alignment and knowledge alignment into the MCL, saving on the additional storage cost of maintaining them separately while maximizing the effect of KAL.

C. Knowledge-Aided Learning

KAL aims at taking advantage of the extra information to adapt or apply in various downstream tasks, such as image–text retrieval [39], visual question answering [40], and text-to-image generation [41]. In the field of image–text retrieval, Wang et al. [42] extracted consensus information from external knowledge and considered statistical co-occurrence correlations among concepts to craft consensus-aware concept representations. Wang et al. [38] utilized instance-level embeddings branch for generating pseudo-online clustering labels as extra knowledge information. Zhang et al. [39] constructed a concept graph to yield the interventional consensus representation for mining the intrinsic associations among concepts. Despite the effectiveness of the above approaches, they all rely on external knowledge for support, which is laborious and time-consuming and pretrained on additional datasets. An alternative approach involves harnessing the intrinsic information within the dataset for learning, leading to reduced training costs. Dong et al. [43] proposed a dual deep encoding network that incorporates a concept space for incorporating the label information from the dataset. Based on this [43], Zhu et al. [44] devised a complementary-aware space alignment technique to harmonize the learning of inherent knowledge with additional information. Unlike the above methods, we introduce the novel KAL framework to facilitate the acquisition of discriminative representations without any external tools or knowledge. The KAL framework comprises a comprehensive learning process, including knowledge initialization, construction, filtration, and alignment operations, which achieves a better focus on the key concepts.

III. Method

A. Overall Architecture

In this section, we delineate our KAMCL method for RSITR, as illustrated in Fig. 3. We begin with introducing the encoding processing of the visual and textual modalities, which includes our proposed HA module for effectively capturing multilevel visual information. Afterward, we delicately elaborate our KAL framework for RSITR, which is incorporated with the MCL techniques for achieving more comprehensive and precise representations. Finally, we develop a two-step training strategy that aims at promoting model convergence and improving overall performance.

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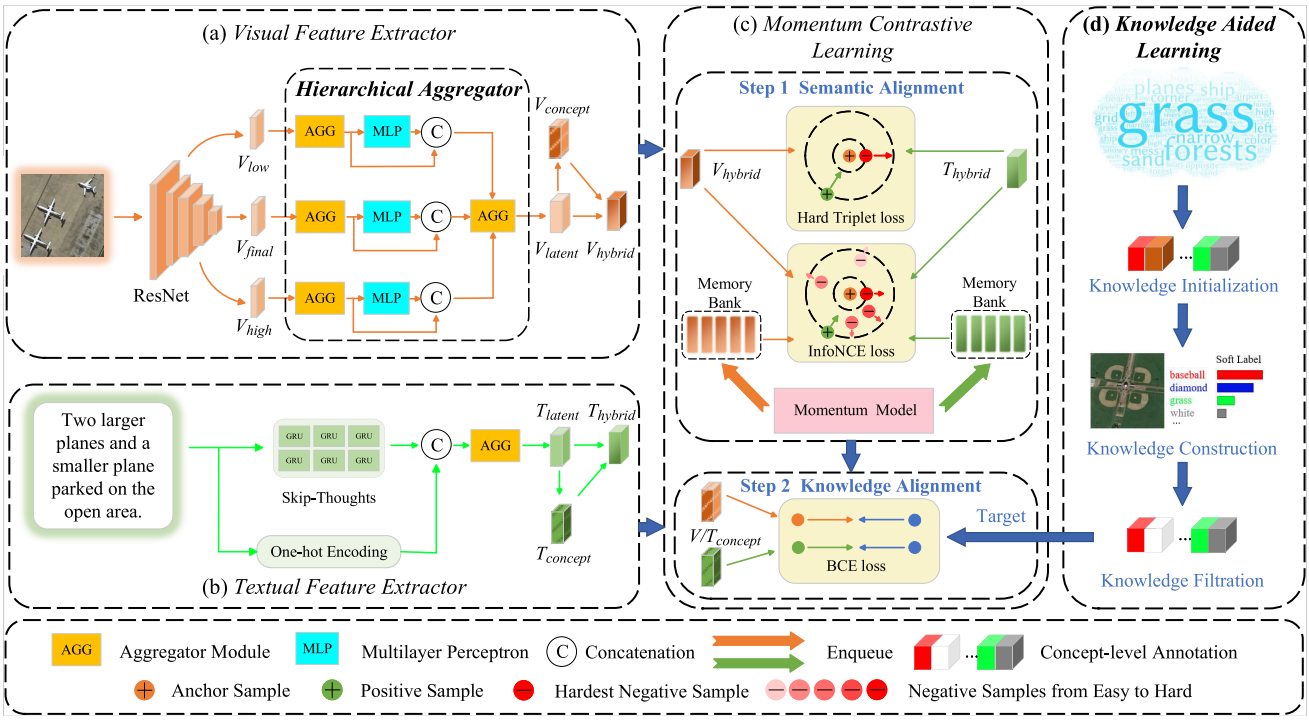


Fig. 3. Overview of our KAMCL approach, which consists of four parts. (a) Visual feature extractor: the ResNet-18 backbone and the proposed HA module are employed as the encoder to obtain visual representations from both the latent level and the concept level. (b) Textual feature extractor: the combination of skip-thoughts and one-hot encoding is applied for the textual representation extraction. (c) MCL: we combine the MoCo training paradigm [29] to realize image–text alignment with two memory banks on the fly. (d) KAL: it includes knowledge initialization, construction, filtration, and alignment operations, to achieve a better focus on the key concepts.

B. Visual Feature Extractor

Following [4], we employ the ResNet-18 [45] as the visual backbone. Specifically, given an input image I, we extract the hierarchical features from four specific hidden layers and the final layer, which are denoted as v_1, v_2, v_3, v_4, and V_{final}, respectively. Subsequently, these features will be fed into our proposed HA module.

1) Hierarchical Aggregator: To obtain multilevel information, we follow Yuan et al. [17] and implement an upsampling strategy on v_2, which is subsequently concatenated with v_1 and subjected to a linear transformation to generate the low-level local feature V_{low}. Similarly, we adopt the same strategy to v_3 and v_4 and generate the high-level salient feature V_{high}. Mathematically, the above operations are defined as follows:

$$V_{\text{low}} = \mathbf{W}_l(\text{Concat}(\text{Upsample}(v_2), v_1)) + b_l \tag{1}$$

$$V_{\text{high}} = \mathbf{W}_h(\text{Concat}(\text{Upsample}(v_4), v_3)) + b_h \tag{2}$$

where $\mathbf{W}_l \in \mathbb{R}^{d_l \times d}$ and $\mathbf{W}_h \in \mathbb{R}^{d_h \times d}$ represent the learnable parameters matrices, d_l, d_h, and d denote the dimensionalities of the original low feature, the original high feature, and the visual representation, respectively, b_l and b_h represent the bias terms.

Our HA module leverages the aforementioned multilevel information. The local information V_{low}, the salient information V_{high}, and the global information V_{final} of ResNet-18 are fed into an aggregator, termed as AGG in this article. Subsequently, the hierarchical features undergo MLP layers respectively. Then they are concatenated to the features from the previous step through a residual connection. We effectively aggregate the information from each part, which is formulated as follows:

$$V_l = \text{Concat}(\text{MLP}(\text{AGG}(V_{\text{low}})), \quad \text{AGG}(V_{\text{low}})) \tag{3}$$

$$V_h = \text{Concat}\big(\text{MLP}\big(\text{AGG}\big(V_{\text{high}}\big)\big), \quad \text{AGG}\big(V_{\text{high}}\big)\big) \tag{4}$$

$$V_f = \text{Concat}(\text{MLP}(\text{AGG}(V_{\text{final}})), \quad \text{AGG}(V_{\text{final}})) \tag{5}$$

where the lower-level feature V_l emphasizes local information V_{low}, the higher-level feature V_h prioritizes salient information V_{high}, and the final feature V_f captures the global information of the entire image V_{final}.

Finally, the final visual feature of the latent space, denoted as V_{latent}, is obtained by passing through the final AGG module, which is formulated as follows:

$$V_{\text{latent}} = \text{AGG}\big(\text{Concat}\big(V_l, V_h, V_f\big)\big). \tag{6}$$

2) AGG Module: We follow the plug-and-play aggregator architecture proposed in [46] as our AGG module. It obtains the aggregated features V_{agg} by automatically learning the weights of the pooling coefficients θ of K-MaxPool [47], which is defined as

$$V_{\text{agg}} = \sum_{k=1}^{N} \theta_k \cdot \max_k \left(\{V_i\}_{i=1}^{N}\right) \tag{7}$$

where $\sum_{k=1}^{N} \theta_k = 1$, N represents the number of feature vectors, which refers to the number of channels during the encoding process, and θ_k represents the coefficient of the kth maximum value among N elements. Specifically, we construct a parameterized generator, which primarily consists of a position encoder based on trigonometric functions and a position

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decoder based on bidirectional gated recurrent unit (Bi-GRU) [48].

a) Position encoder: It applies a trigonometric function to encode each position index k to a dense vector p_k. The position encoder is widely employed in Transformer [49], and its structure is defined as follows:

$$p_k^i = \begin{cases} \sin(w_j, k), & \text{when } i = 2j \\ \cos(w_j, k), & \text{when } i = 2j+1 \end{cases} \tag{8}$$

where $w_j = (1/10000^{2j/d_k})$ and d_k is the dimensionality of positional encoding.

b) Position decoder: We put the obtained dense vector p_k into a position decoder and output the pooled coefficient θ_k. In particular, the position decoder includes a Bi-GRU [48] and an MLP layer

$$\{h_k\}_{k=1}^N = \text{BiGRU}(\{p_k\}_{k=1}^N) \tag{9}$$

$$\theta_k = \text{MLP}(h_k) \tag{10}$$

where h_k is the output of Bi-GRU [48] at position k. Finally, we aggregate features into the representation space.

Inspired by [43] and [44], we obtain the visual concept feature V_{concept} by passing the latent feature V_{latent} through linear and sigmoid layers, which are employed in our proposed KAL framework. Specifically, the concept space aids in predicting concept labels for different modalities, thereby converting the retrieval problem into a multilabel classification task, which further aids in the retrieval task. We obtain the visual feature V_{hybrid} by a simple addition of both, which unifies latent and concept representations

$$V_{\text{concept}} = \sigma(\mathbf{W}_c V_{\text{latent}} + b_c) \tag{11}$$

$$V_{\text{hybrid}} = V_{\text{latent}} + V_{\text{concept}} \tag{12}$$

where $\mathbf{W}_c \in \mathbb{R}^{d_c \times d}$ represents the learnable parameters matrices, d_c denotes the dimensionality of the concept space, b_c represents the bias terms, and σ is the sigmoid function.

C. Textual Feature Extractor

For the textual feature extractor, we employ skip-thoughts [50] as the backbone of the text modality. At the same time, we utilize one-hot encoding to obtain the corresponding vector T_E for a given sentence T and concatenate it with the text feature T_{GRU} as the input of AGG module to obtain the latent-level text feature T_{latent}, which are formulated as

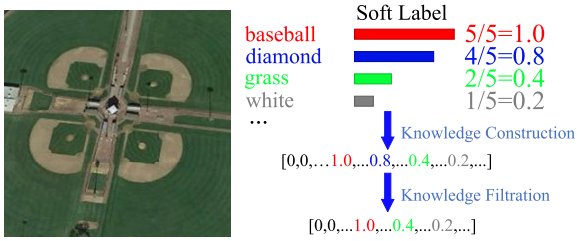
$$T_{\text{GRU}} = \text{MLP}(\text{GRU}(T_E)) \tag{13}$$

$$T_{\text{latent}} = \text{AGG}(\text{Concat}(T_E, T_{\text{GRU}})). \tag{14}$$

Similar to the visual branch, we incorporate a linear layer and a sigmoid layer to generate the concept feature T_{concept} for the textual representation. By aggregating it with the latent feature T_{latent}, we obtain the hybrid feature T_{hybrid}. The formulations of the aforementioned process are expressed as follows:

$$T_{\text{concept}} = \sigma(\mathbf{W}_c T_{\text{latent}} + b_c) \tag{15}$$

$$T_{\text{hybrid}} = T_{\text{latent}} + T_{\text{concept}} \tag{16}$$



Four baseball diamonds next to a white building .
There is a trail leading to this baseball field .
There are four baseball diamonds in the grass .
Four baseball diamonds are distributed around the two roads .
Four baseball diamonds are surrounded by grass .

Fig. 4. Specific processes of the KAL framework. 1) Knowledge initialization: the high-frequency words within the dataset are selected as the concept word list in advance. 2) Knowledge construction: the word frequency of the five sentences to a single image is counted as a soft label. 3) Knowledge filtration: the soft label is compared with the concept word list to filter out words that do not exist in the list and cumulatively transform them into concept-level annotations. 4) Knowledge alignment: concept-level annotations are considered as the targets of the features V_{concept} and T_{concept} and aligned with them.

where $\mathbf{W}_c \in \mathbb{R}^{d_c \times d}$ represents the learnable parameters matrices, d_c denotes the dimensionality of the concept space, b_c represent the bias terms, and σ is the sigmoid function.

D. Knowledge-Aided Learning

In this section, we describe our proposed KAL framework, which aims at learning more discriminative representations with the help of inherent knowledge within the dataset. Note that our KAL framework does not rely on any external knowledge or tools.

1) Knowledge Initialization: We begin with introducing the knowledge initialization method, which involves performing word frequency statistics on all text within the dataset. Partially, we select the top k words with high frequency across the entire dataset. Note that high-frequency words such as "a," "the," and "of" are excluded from the list and are replaced as "<UNK>" in the dictionary. We aim to enhance the meaningfulness of the concept space and ensure its consistency with the multilabel attributes of the corresponding image. The formulation of the knowledge initialization operation is summarized as follows:

$$C_{\text{vocabulary}} = \text{Top}_k\{\text{Frequency}(T_{\text{dataset}} - < \text{UNK} >)\} \tag{17}$$

where T_{dataset} represents all the words within the dataset, Frequency means the frequency statistics of all words, which are then sorted in descending order, and $C_{\text{vocabulary}}$ is the key concept list of the entire dataset.

2) Knowledge Construction: We construct the knowledge on each matched image–text pair in advance. Specifically, in this process, the frequency of each word in the sentences corresponding to the image are counted as a soft label, which is formulated as follows:

$$L_{\text{word}}, L_{\text{soft}} = \text{Frequency}(T_{\text{sentence}} - < \text{UNK} >) \tag{18}$$

where T_{sentence} represents all corresponding sentences to an image, Frequency means the frequency statistics of all words,

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which are then sorted in descending order, L_{word} means the label words in sentence-level statistics, and L_{soft} is the corresponding soft label, which is calculated as depicted in Fig. 4. For example, the word "baseball" appears in all five sentences, so its L_{soft} value is $5/5 = 1.0$. The word "diamond" appears four times in the five sentences, so its L_{soft} value is $4/5 = 0.8$. Finally, We obtain all the label words and corresponding soft label values by the above statistics. We refer to this process as knowledge construction.

3) Knowledge Filtration: The high-frequency words L_{soft} are then compared with the concept word list $C_{\text{vocabulary}}$ for knowledge filtration, which aims at extracting the relevant concept-level annotation. The formulation for it is as follows:

$$A_{\text{concept}} = \bigcap_{i=1}^{k} \{A_i = L_{\text{soft}} | L_{\text{word}} \in C_{\text{vocabulary}}\} \tag{19}$$

where A_i represents the annotation at position i and A_{concept} represents the concept-level annotations. As depicted in Fig 4, the word "diamond" is filtered for its absence in the key concept list $C_{\text{vocabulary}}$, despite its high frequency within the sentences.

Then, we utilize the obtained concept-level annotations as the target for the knowledge alignment. By optimizing the model to bring the concept representations of images and text closer to the target, we aim to achieve knowledge alignment, which helps in learning more meaningful and effective semantic space for image–text retrieval.

4) Knowledge Alignment: To achieve more precise representations, we employ the prediction of key concepts in both modalities to facilitate learning in a shared embedding space. Specifically, we utilize a binary cross-entropy (BCE) loss [51], with concept-level annotations A_{concept} as its labels (denoted as "y"), to guide the convergence of the concept spaces for images and text. Here is the calculation method for a given image–text pair (I, T) and the corresponding real label y

$$\mathcal{L}_{\text{bce_}I}(I, y) = -\frac{1}{K}\sum_{i=1}^{K}\left[y_i \log(g(I)_i) + (1 - y_i)\log(1 - g(I)_i)\right] \tag{20}$$

$$\mathcal{L}_{\text{bce_}T}(T, y) = -\frac{1}{K}\sum_{i=1}^{K}\left[y_i \log(g(T)_i) + (1 - y_i)\log(1 - g(T)_i)\right] \tag{21}$$

where $g(I)$ and $g(T)$ denote the visual and textual representations in the concept space, which predict the label output at the corresponding ith position.

We employ the knowledge alignment for both text and images, summing them as the loss function of concept space

$$\mathcal{L}_{\text{bce}} = \mathcal{L}_{\text{bce_}I} + \mathcal{L}_{\text{bce_}T}. \tag{22}$$

By employing BCE loss for knowledge alignment, the model aims to minimize the discrepancy between the predicted probabilities and the real labels, thereby aligning the concept spaces of the visual and textual modalities. This mechanism improves the model's attention to prominent concepts and achieves favorable retrieval results.

E. Momentum Contrastive Learning

Existing traditional training mechanisms [4], [17], [26] rely solely on samples in each mini-batch, which limits the scale of negative samples and constraints the representation capacity. Inspired by the pioneering work in unsupervised visual representation learning MoCo [29] and its various downstream works [21], [29], [52], [53], we incorporate our proposed KAL framework with MoCo [29] for maximizing its effect and learning precise and discriminative representations. By integrating the MCL, the model mines the information of negative samples on a larger scale. We implement the method by utilizing two memory banks B_t and B_v, which serve as dynamic queues for storing negative samples. Additionally, we construct key encoders θ_k^t and θ_k^v, which are updated based on momentum. This ensures the consistency of representations obtained by the key encoder.

1) Textual Memory Bank: We construct the textual memory bank B_t with the size K_t as a dynamic queue to store the textual representation T_{hybrid}. During each training iteration, the key textual representation of the current batch is enqueued by the key textual encoder θ_k^t, while the oldest batch's data is dequeued from the memory bank. The key textual representation is then utilized to compute the similarity with the query visual encoder of the current batch for generating negative pairs $\sum_{i=1}^{K_t} s_i^{vt^-}$.

2) Visual Memory Bank: Similarly, the visual memory bank B_v with the size K_v for the visual modality is applied as a dynamic queue to store the visual key representation V_{hybrid}. During training iterations, the key visual representation of the current batch is enqueued by the key visual encoder θ_k^v, while dequeuing the oldest representations from the memory bank. Subsequently, the key visual representation is utilized to calculate similarity with the query textual encoder of the current batch for the negative pairs $\sum_{i=1}^{K_v} s_i^{tv^-}$.

Then, we formulate the momentum update method for θ_k^t and θ_k^v as

$$\theta_k^t = m\theta_k^t + (1 - m)\theta_q^t \tag{23}$$

$$\theta_k^v = m\theta_k^v + (1 - m)\theta_q^v \tag{24}$$

where $m \in [0, 1)$ is a momentum coefficient, which is a relatively large value and set to $m = 0.99$ in this article. Finally, though the key representations in memory banks are obtained by encoders under different parameters, the updates to their representations will be smooth, which ensures the continuity of representations.

InfoNCE Loss: To leverage a larger scale of negative samples, we employ InfoNCE loss [54] as one of the objective functions for semantic alignment. Initially, we compute the positive similarities s^{vt^+} and s^{tv^+} between the query textual/visual representation and the key visual/textual representations in the current batch. Subsequently, we calculate the negative similarities $\sum_{i=1}^{K_t} s_i^{vt^-}$ and $\sum_{i=1}^{K_v} s_i^{tv^-}$ between the query textual/visual representations in the current batch and the negative visual/textual representations stored in the memory bank. Subsequently, we get $\sum_{i=1}^{1+K_t} s_i^{vt}$ and $\sum_{i=1}^{1+K_v} s_i^{tv}$ by concatenating the aforementioned positive and negative similarities. Then, we calculate the loss by the following

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formula:

$$\mathcal{L}_{\text{info}} = -\frac{1}{N}\sum_{m=1}^{N}\left(\log\frac{\exp\left(s^{vt^+}/\gamma\right)}{\sum_{i=1}^{1+K_t}\exp\left(s_i^{vt}/\gamma\right)} + \log\frac{\exp\left(s^{tv^+}/\gamma\right)}{\sum_{i=1}^{1+K_v}\exp\left(s_i^{tv}/\gamma\right)}\right) \tag{25}$$

where γ is a temperature hyperparameter and set to 0.07, K_t and K_v represent the size of the textual and visual memory banks, respectively, and N is the batch size.

F. Training Strategy

We delicately design a training strategy to train our model, which effectively imparts more meaningful label information to the concept space upon the foundation of semantic space alignment.

Triplet Loss: In addition to the aforementioned BCE loss and InfoNCE loss, we also employ the popular triplet loss [7] with the hardest negative sample as another training objective, which is defined as follows:

$$\mathcal{L}_{\text{tri}}(v,t) = \max_{t^-}\left[\alpha + s^{vt^-} - s^{vt}\right]_+ + \max_{v^-}\left[\alpha + s^{tv^-} - s^{vt}\right]_+ \tag{26}$$

where α represents the margin value, s^{vt} are the similarity of the matched sample pairs, and s^{vt^-} and s^{tv^-} are the mismatched image–text pairs. Specifically, triplet loss only focuses on the hardest negative sample, while the InfoNCE loss pays more attention to a larger scale of negative pairs, and the BCE loss emphasizes the significance of the key concepts in distinguishing the analogous descriptions. Additionally, we observe that directly employing the BCE loss from the outset is not conducive to achieving a well-learned embedding space, as the emphasis of the objective functions differs in semantic and knowledge alignments. A more detailed explanation will be provided in the experimental section.

Considering the different training objectives possess different optimization characteristics, we propose a novel two-step training strategy. Specifically, we divide the whole training process into two steps, where the first step adopts the triplet loss and the InfoNCE loss for the semantic alignment. The second step additionally incorporates the BCE loss to optimize the concept space for knowledge alignment. Mathematically, the specific training objective of the two steps are defined as follows:

$$\mathcal{L}_{\text{step1}} = \lambda_1\mathcal{L}_{\text{tri}} + \lambda_2\mathcal{L}_{\text{info}}, \tag{27}$$

$$\mathcal{L}_{\text{step2}} = \lambda_1\mathcal{L}_{\text{tri}} + \lambda_2\mathcal{L}_{\text{info}} + \lambda_3\mathcal{L}_{\text{bce}}. \tag{28}$$

By incorporating BCE loss in the second step, we effectively guide the model to pay attention to the concept-level information without compromising the importance of the semantic space.

This strategy ensures a balanced training process that considers both the semantic alignment and the knowledge alignment, allowing the model to converge efficiently. The experimental results indicate that this two-step training strategy greatly accelerates the convergence rate of the model and achieves better results.

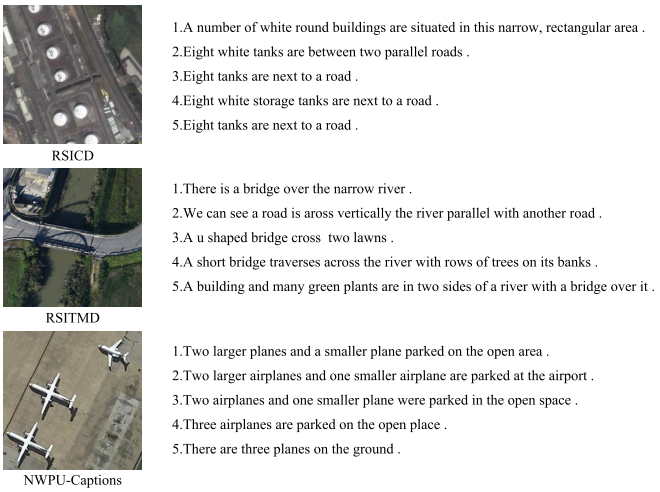


Fig. 5. Samples from the RSICD, RSITMD, and NWPU-Captions datasets.

IV. Experiments Results and Analysis

A. Dataset and Evaluation Metrics

We verify the effectiveness of our method by conducting experiments on the RSICD [15], RSITMD [17], and NWPU-Captions [22] datasets. RSICD contains 10 921 images with the size of 224 × 224, which is the most commonly employed dataset for RSITR. RSITMD contains 4743 images with the size of 256 × 256. In addition to the above two widely employed datasets, we additionally validate the NWPU-Captions dataset on RSITR which contains 31 500 images with the size of 256 × 256. Each image corresponds to five sentences as shown in Fig. 5. We split these three datasets into 80%, 10%, and 10% for the training set, validation set, and testing set following the protocol of [17].

We employ the popular Recall at k ($R@k$, k=1,5,10) and meanR (mR) [55] as the evaluation metrics to assess the retrieval performance on both the sentence retrieval and image retrieval. Specifically, $R@k$ aims to calculate the proportion of ground truth in the top k samples. mR is the average value of all six $R@k$, which illustrates the overall retrieval performance. We select the model with the highest mR on the validation set for the test.

B. Implementation Details

Following [17] and [4], we employ identical data augmentation techniques for fair comparison. Specifically, all images are resized into 256 × 256 pixels and performed data augmentations of RandomRotation, RandomCrop, and Normalize to enhance the model's robustness. The dimensionalities of the latent and concept embedding space for both modalities are set to 512. We train the model for 20 epochs with a batch size of 100 and employ Adam [56] as our optimizer. The initial learning rate is 2e-4 and decays by a factor of 0.5 every ten epochs. After ten epochs, we add BCE loss into the objective function. To verify the stability of our model, we report the average result by repeating it five times under the same settings. All experiments are implemented with PyTorch and trained with a single NVIDIA GeForce RTX 4090 GPU.

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TABLE I
COMPARISONS OF RETRIEVAL PERFORMANCE ON RSICD AND RSITMD TESTSET

Approach	Backbone	Extra Tools	RSICD							RSITMD						
			Sentence Retrieval			Image Retrieval			mR	Sentence Retrieval			Image Retrieval			mR
			R@1	R@5	R@10	R@1	R@5	R@10		R@1	R@5	R@10	R@1	R@5	R@10	
VSE++$_{BMVC'18}$ [7]	ResNet-18	-	3.38	9.51	17.46	2.82	11.32	18.10	10.43	10.38	27.65	39.6	7.79	24.87	38.67	24.83
SCAN t2i$_{ECCV'18}$ [8]	ResNet-18	Faster R-CNN [57]	4.39	10.90	17.64	3.91	16.20	26.49	13.25	10.18	28.53	38.49	10.1	28.98	43.53	26.64
SCAN i2t$_{ECCV'18}$ [8]	ResNet-18	Faster R-CNN [57]	5.85	12.89	19.84	3.71	16.40	26.73	14.23	11.06	25.88	39.38	9.82	29.38	42.12	26.28
CAMP-triplet$_{ICCV'19}$ [58]	ResNet-18	Faster R-CNN [57]	5.12	12.89	21.12	4.15	15.23	27.81	14.39	11.73	26.99	38.05	8.27	27.79	44.34	26.20
CAMP-bce$_{ICCV'19}$ [58]	ResNet-18	Faster R-CNN [57]	4.20	10.24	15.45	2.72	12.76	22.89	11.38	9.07	23.01	33.19	5.22	23.32	38.36	22.03
MTFN$_{ACMMM'19}$ [59]	ResNet-18	Faster R-CNN [57]	5.02	12.52	19.74	4.90	17.17	29.49	14.81	10.40	27.65	36.28	9.96	31.37	45.84	26.92
LW-MCR(b)$_{TGRS'21}$ [26]	ResNet-18	-	4.57	13.71	20.11	4.02	16.47	28.23	14.52	9.07	22.79	38.05	6.11	27.74	49.56	25.55
LW-MCR(d)$_{TGRS'21}$ [26]	ResNet-18	-	3.29	12.52	19.93	4.66	17.51	30.02	14.66	10.18	28.98	39.82	7.79	30.18	49.78	27.79
AMFMN-soft$_{TGRS'22}$ [17]	ResNet-18	-	5.05	14.53	21.57	5.05	19.74	31.04	16.02	11.06	25.88	39.82	9.82	33.94	51.90	28.74
AMFMN-fusion$_{TGRS'22}$ [17]	ResNet-18	-	5.39	15.08	23.40	4.90	18.28	31.44	16.42	11.06	29.20	38.72	9.96	34.03	52.96	29.32
AMFMN-sim$_{TGRS'22}$ [17]	ResNet-18	-	5.21	14.72	21.57	4.08	17.00	30.60	15.53	10.63	24.78	41.81	11.51	34.69	54.87	29.72
GaLR w/o MR$_{TGRS'22}$ [4]	ResNet-18	Ppyolo [60]	6.50	18.91	29.70	5.11	19.57	31.92	18.62	13.05	30.09	42.70	10.47	36.34	53.35	31.00
GaLR with MR$_{TGRS'22}$ [4]	ResNet-18	Ppyolo [60]	6.59	19.85	31.04	4.69	19.48	32.13	18.96	14.82	31.64	42.48	11.15	36.68	51.68	31.41
KCR$_{IJCAI'22}$ [28]	ResNet-101	-	5.95	19.59	29.58	5.40	22.44	37.36	19.89	-	-	-	-	-	-	-
SWAN$_{ICMR'23}$ [16]	ResNet-50	Glove [61]	7.41	20.13	30.86	5.56	22.26	37.41	20.61	13.35	32.15	46.90	11.24	40.4	**60.60**	34.11
KAMCL(Ours)	ResNet-18	-	9.24	22.51	36.23	6.70	24.13	39.14	23.26	**17.06**	35.84	48.45	13.98	**42.35**	59.51	**36.19**
KAMCL(Ours)	ResNet-50	-	**12.08**	**27.26**	**38.70**	**8.65**	**27.43**	**42.51**	**26.10**	16.51	**36.28**	**49.12**	13.50	42.15	59.32	36.14
KAMCL(Ours)	ResNet-101	-	11.99	27.17	38.33	7.78	25.23	40.02	25.08	16.81	34.96	47.12	**14.60**	41.86	59.86	35.87

TABLE II
COMPARISONS OF RETRIEVAL PERFORMANCE ON NWPU-CAPTIONS TESTSET. RESULTS ARE REPLICATED BY THE OPEN-SOURCE CODE

Approach	Backbone	Sentence Retrieval			Image Retrieval			mR
		R@1	R@5	R@10	R@1	R@5	R@10	
VSE++$_{BMVC'18}$ [7]	ResNet-18	4.84	12.89	20.94	4.38	13.61	24.12	13.46
AMFMN-soft$_{TGRS'22}$ [17]	ResNet-18	6.62	24.82	40.05	5.97	22.99	38.09	23.09
AMFMN-fusion$_{TGRS'22}$ [17]	ResNet-18	11.49	38.75	57.73	8.63	30.25	46.48	32.22
AMFMN-sim $_{TGRS'22}$ [17]	ResNet-18	9.13	34.16	52.32	23.41	28.41	44.80	29.44
KAMCL(Ours)	ResNet-18	20.14	54.10	72.05	11.51	35.33	51.33	40.75
KAMCL(Ours)	ResNet-50	**21.02**	**57.33**	**74.41**	**12.74**	**38.03**	**53.90**	**42.90**
KAMCL(Ours)	ResNet-101	20.57	54.76	71.55	12.25	36.80	52.95	41.48

C. Comparisons With the SOTA Methods

We compare our proposed KAMCL method with the existing state-of-the-art methods, including visual-semantic embeddings (VSE++) [7], stacked cross attention network (SCAN) [8], cross-modal adaptive message passing (CAMP) [58], multi-modal tensor fusion network (MTFN) [59], asymmetric multimodal feature matching network (AMFMN) [17], lightweight multi-scale cross-modal retrieval (LW-MCR) [26], global and local information (GaLR) [4], knowledge-aware cross-modal retrieval (KCR) [28], and scene-aware aggregation network (SWAN) [16]. Note that for the sake of fair comparison, we employ identical data augmentation techniques. In our experiments, we also investigate the impact of different backbones on the performance of our model, including ResNet-18, ResNet-50, and ResNet-101.

The experimental results of RSICD, RSITMD, and NWPU-Captions datasets are shown in Tables I and II. In addition, model configuration information, including the network backbones and extra tools used for visual and textual extraction, is provided. Unless specifically stated, we employ ResNet-18 for fair comparisons with other methods. From the two tables, we have several observations and conclusions as follows.

1) For the RSICD dataset, our KAMCL method notably outperforms all the competing methods across various evaluation metrics. Specifically, compared with SWAN [16], the previous best method, our KAMCL exceeds it by 2.65% on mR, despite our utilization of a more lightweight backbone. For the RSITMD dataset, our KAMCL method achieves an accuracy of 36.19% on mR, outperforming the state-of-the-art method SWAN by 2.08%. Additionally, compared with GaLR [4] and SWAN [16], our method does not require any additional bounding box assistance or external common knowledge. This makes our method more convenient for the training and testing process.
2) For the NWPU-Captions dataset, we replicate the results by employing their released codes without modifying any setting. It is worth noting that we do not replicate GaLR [4] due to the requirement of some indispensable details of utilizing Baidu's paddle platform [60] and its specific bounding box-related operations to get adjacency matrix and representation matrix. However, our method, along with AMFMN, does not rely on any extra object detector. The results on NWPU-Captions demonstrate that our method exhibits a striking improvement of 8.53% over the best-performing model, with the mR of 40.75%. Overall, these results showcase the effectiveness and superiority of our method across different datasets.
3) In comparison with traditional retrieval methods like VSE++ [7], SCAN [8], CAMP [58], and MTFN [59], remote-sensing specific methods frequently demonstrate superior performance. KAMCL outperforms the MTFN method by 8.45 % on mR on the RSICD dataset. This is attributed to the fact that remote-sensing-specific methods take into account the unique characteristics of the remote-sensing field and place a strong emphasis on addressing its distinct challenges. Remote-sensing data often presents more analogous descriptions and image morphologies compared with natural retrieval tasks, leading to an abundance of hard samples. KAMCL along with other remote-sensing-specific methods excels in tackling the learning of highly similar remote-sensing images, thereby achieving superior retrieval performance.
4) The results of different backbones demonstrate that the overall performance of ResNet-50 surpasses those of

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TABLE III
ABLATION EXPERIMENTS FOR DIFFERENT STRUCTURES OF KAMCL ON RSICD TESTSET

Model	HA	MCL	KAL	Sentence Retrieval R@1	R@5	R@10	Image Retrieval R@1	R@5	R@10	mR
1				6.12	19.76	30.83	4.63	20.09	33.54	19.16
2	✓			7.68	21.13	33.12	6.33	22.87	37.62	21.46
3	✓	✓		9.13	22.43	34.62	6.33	23.22	37.24	22.16
4	✓		✓	8.32	22.32	35.59	6.13	22.37	38.26	22.17
5	✓	✓	✓	**9.24**	**22.51**	**36.23**	**6.70**	**24.13**	**39.14**	**23.26**

TABLE IV
COMPARISON OF DIFFERENT AGGREGATION MODULE ON THE RSICD TESTSET

Model	Approach	Sentence Retrieval R@1	R@5	R@10	Image Retrieval R@1	R@5	R@10	mR
1	None	6.12	19.76	30.83	4.63	20.09	33.54	19.16
2	Mean	7.21	20.12	31.20	5.56	21.45	34.11	19.94
3	Max	6.49	21.32	31.47	5.25	21.46	34.98	20.16
4	MVSA [4]	6.86	19.21	29.27	5.05	20.73	34.56	19.28
5	Self-attention [49]	7.28	**21.19**	**33.15**	5.60	22.36	36.61	20.94
6	HA(Ours)	**7.68**	21.13	33.12	**6.33**	**22.87**	**37.62**	**21.46**

TABLE V
ABLATION STUDY ON THE DIMENSIONALITY OF CONCEPT SPACE TO INVESTIGATE THE CONTRIBUTIONS OF KAL ON THE RSICD TESTSET

Concept Embedding	Sentence Retrieval R@1	R@5	R@10	Image Retrieval R@1	R@5	R@10	mR
32	8.42	21.59	33.03	5.97	22.73	37.38	21.52
64	9.33	23.78	35.15	**7.02**	22.96	38.86	22.85
128	9.70	24.06	35.22	6.99	23.11	37.79	22.81
256	**9.24**	24.06	35.68	6.40	23.37	38.83	22.93
512	**9.24**	22.51	**36.23**	6.70	**24.13**	**39.14**	**23.26**
1,024	8.33	**24.42**	35.86	6.51	23.60	39.03	22.96

both ResNet-18 and ResNet-101 with improvements of 1.02% on the RSICD dataset and 1.42% on the NWPU-Captions dataset. Despite the slight deterioration of the RSITMD dataset, this could be attributed to the limited scale of the dataset, resulting in overfitting despite its better performance on the validation set. This finding is consistent with the observations highlighted in [45], indicating that an overly deep ResNet network fails to improve visual representations.

D. Ablation Studies

To examine the effectiveness of our proposed KAMCL, we conduct extensive ablation studies with the backbone of ResNet-18. These studies involve evaluating various aspects of the KAMCL framework, such as different configurations, aggregation modules, concept space dimensionality, memory bank sizes, and training objectives. Through these extensive ablation studies, we aim to gain insights into the individual contributions and impact of each component.

1) Ablation Studies of Structures: As shown in Table III, our ablation studies are conducted progressively by gradually adding the proposed components for comparison. In Model 1, we simply add the output of each layer of ResNet-18, while in Model 2, we employ our proposed HA module for aggregation. Subsequently, a notable increase of 2.3% on mR demonstrates that our designed module comprehends multilevel information more effectively. Upon incorporating MCL into our model, as demonstrated in Model 3, we observe a further comprehensive improvement compared with Model 2, with an increase of 0.7% on mR. Transitioning to Model 4, where we implement the KAL, an increase of 0.71% on mR is observed when contrasted with Model 2. Ultimately, by simultaneously applying all three modules in Model 5, a total 4.1% increase in mR showcases the striking performance of our model. These ablation experiments validate the roles of our designed modules in enhancing cross-modal retrieval performance.

2) Aggregation Module: To validate the effectiveness of our proposed HA module, we conduct comparative experiments by adopting the original triplet loss as the objective function for fair comparison. As shown in Table IV, we compare our proposed HA module with several aggregation modules, including mean pooling, max-pooling, multi-level visual self-attention (MVSA) module [4], and self-attention [42], [62].

In Model 1, we directly utilize the original features of ResNet-18 without employing the aggregation module for an intuitive comparison. For a fair comparison, we incorporate multilevel information of the visual backbone by a simple mean or max-pooling in Models 2 and 3. The result indicates that multilevel information is beneficial for learning better representation. Additionally, we employ the MVSA module [17] in Model 4 to aggregate features, as designed in GaLR [4] and AMFMN [17], which utilizes the ResNet-18's four-layered structure for residual fusion and incorporates an attention mechanism to generate attention maps that highlight key areas for aggregating visual features. In Model 5, we adopt the currently prevalent self-attention mechanism [49] to aggregate the visual representations.

Compared with Model 1, our proposed HA module exhibits a 2.23% improvement in mR, which confirms that our HA module effectively captures the underlying structure information and integrates it into the upper features. Furthermore, compared with the MVSA [17] in Model 4, our designed module demonstrates a 2.18% improvement in accuracy. In addition, though the result of the self-attention mechanism shows comparable performance in sentence retrieval, HA shows superior performance in image retrieval.

3) Quantitative Analysis of the Dimensionality of the Concept Space: As shown in Table V, we also conduct further research on the dimensionality of the concept space, which signifies the number of key concepts being considered.

From Table V, we observe that when the concept word list exceeds 64, the differences are not significant. However, there is still a noticeable trend indicating that higher numbers of concepts lead to better retrieval performance. The experimental outcomes manifest that employing the concept dimensionality of 512 results in significantly enhanced performance. This observation may be attributed to the dataset's vocabulary, with RSICD, RSITMD, and NWPU-Captions boasting vocabularies comprising 1226, 1350, and 1275 words, respectively. Consequently, the adoption of excessively extensive concepts might lead to a dispersion of focus. Thus, the selection of an

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TABLE VI

ABLATION STUDY ON THE EFFECTIVENESS OF MEMORY BANK TO INVESTIGATE THE CONTRIBUTIONS OF MCL ON THE NWPU-CAPTIONS TESTSET

Batch Size	Memory Bank		Sentence Retrieval			Image Retrieval			mR
	Kv	Kt	R@1	R@5	R@10	R@1	R@5	R@10	
50	0	0	13.11	42.22	61.21	9.97	32.53	48.77	34.63
100	0	0	14.98	44.22	63.02	9.75	32.88	49.33	35.70
500	0	0	15.12	45.19	64.13	9.63	33.96	49.25	36.21
100	500	0	16.54	49.52	67.30	11.01	34.34	49.87	38.09
100	0	500	17.21	50.98	67.92	10.76	33.78	49.24	38.31
100	500	500	17.14	50.70	68.70	10.91	34.12	50.09	38.61
100	5,000	5,000	18.09	51.05	68.06	10.89	34.54	50.20	38.80
100	10,000	10,000	19.30	51.52	69.24	10.93	34.29	50.36	39.27
100	15,000	15,000	19.97	52.03	69.65	11.36	35.17	51.15	39.89
100	**20,000**	**20,000**	**20.14**	**54.10**	**72.05**	**11.51**	**35.33**	**51.33**	**40.75**
100	25,000	25,000	19.84	53.34	72.03	11.13	35.22	51.30	40.47

appropriate concept dimensionality, as exemplified here with 512, appears to be judicious.

4) Memory Bank: To further investigate the effectiveness of the MCL, we conduct ablation experiments to explore the impact of the memory bank on the NWPU-Captions dataset, for its larger volume of data. As presented in Table VI in lines 1–3, we primarily focus on investigating the impact of batch size. The results show that a larger batch size results in improved model performance and a more stable training process.

Furthermore, in lines 4–6, we delve into the influence of the visual memory bank and the textual memory bank. Regardless of whether utilized individually, both memory banks contribute to better performance through a large scale of negative samples. Remarkably, the combination of these two yields favorable results, facilitating the simultaneous enhancement of visual and textual representations. Even a memory bank as small as 500 is capable of significantly enhancing the model's performance.

To further explore the impact of the memory bank size, we vary the size of K_v and K_t from 5000 to 25 000 and evaluate $R@k$ and mR metrics. As the memory bank increases, the model performance gradually improves, as it allows for expanding the scale of negative samples. However, we observe that once the size of the memory bank exceeds a certain threshold, the model performance slightly decreases. This could be attributed to the notion that having a larger number of negative samples is not conducive to effective learning, as it has the potential to create confusion among samples that share closer semantic information. Consequently, these samples might interfere with the learning process of the model. Based on our experimental results, we observe that setting the size of the memory bank at around 70%–80% of the dataset size yields the best performance.

5) Objective Function: To further investigate the impact of different training objectives, we conduct the experiments progressively by gradually adding the different types of objectives to the RSICD dataset.

In our experiment, we divide triplet loss into the original version and one of the hardest negative sample mining. We compare the impact of the above two versions in Models 1 and 2. Specifically, we observe that employing hard triplet loss sometimes yields relatively good performance. As shown in Table VII, Model 1 with hard triplet sometimes achieves a performance of 22.71% (19.58%+3.13%) on mR. However, its training process has become volatile, rendering it less amenable to convergence. Consequently, the model's results also lack stability compared with the original triplet loss. This could be attributed to the fact that hard triplet loss focuses solely on the most challenging sample within a batch, which is highly influenced by batch size and sampling strategy. Henceforth, if the selection of the hardest negative sample proves suboptimal, the model may fail to attain precise representations, potentially leading to a deviation in retrieval. Employing the hard triplet loss might enhance the model's discriminative capabilities. However, ensuring its stability has emerged as a central challenge that we must address.

TABLE VII

ABLATION STUDY ON DIFFERENT COMBINATIONS OF OBJECTIVE FUNCTIONS ON THE RSICD TESTSET. NOTE THAT "H," "T," "I," AND "B" DENOTE HARD TRIPLET, TRIPLET, INFONCE, AND BCE RESPECTIVELY

Model	Approach	Sentence Retrieval			Image Retrieval			mR
		R@1	R@5	R@10	R@1	R@5	R@10	
1	H	7.93±2.34	20.37±2.63	31.44±3.72	5.58±1.56	19.75±4.09	32.39±4.41	19.58±3.13
2	T	7.68±0.31	21.13±0.51	33.12±0.76	6.33±0.10	22.87±0.25	37.62±0.34	21.46±0.46
3	I	7.19±0.24	19.73±0.61	31.23±0.58	4.32±0.08	18.45±0.23	33.15±0.49	19.01±0.37
4	T+I	8.48±0.20	22.43±0.25	35.28±0.44	6.31±0.11	22.63±0.36	38.31±0.47	22.32±0.29
5	H+I	9.10±0.25	22.12±0.34	35.62±0.51	6.33±0.18	23.22±0.54	38.24±0.78	22.43±0.33
6	**H+I+B**	**9.24±0.47**	**22.51±0.40**	**36.23±0.11**	**6.70±0.38**	**24.13±1.07**	**39.14±0.13**	**23.26±0.33**

Therefore, we explore the possibility of employing the hardest negative sample mining to focus on extremely similar samples while also comparing the information from more common negative samples to achieve model stability. To achieve this, we introduce the momentum model and InfoNCE loss [54], directing the model's focus toward a substantial number of common samples. In Models 3, 4, and 5, we demonstrate the employment of InfoNCE loss alone, in combination with original triplet loss, and combination with the hard triplet loss. The results showcase that InfoNCE loss under the momentum model exhibits good stability by considering a large number of negative samples and effectively learning the dataset's characteristics. When combined with the hard triplet loss, the model could focus on the relationships among the most difficult-to-distinguish mismatched image–text pairs, while also comparing a substantial number of ordinary mismatched image–text ones. This combination leads to an extremely stable model that is easy to converge, thereby accelerating the convergence rate. In Model 6, we introduce the BCE loss of the KAL mechanism based on the optimal performance from the fifth set of experiments. The experimental results demonstrate that this module further improves the retrieval performance, thereby validating the effectiveness of our method.

6) Sensitivity Analysis of Loss Function Weights: To delve deeper into the relationship between the required weight hyperparameters for these three loss functions, we conduct several ablation experiments. Our method divides the training process into two steps, allowing us to adjust the weight of the loss function separately in each step to find the optimal hyperparameters. As shown in Fig. 6(a), during step one, we initially set the weight for the hard triplet loss λ_1 to 1 and

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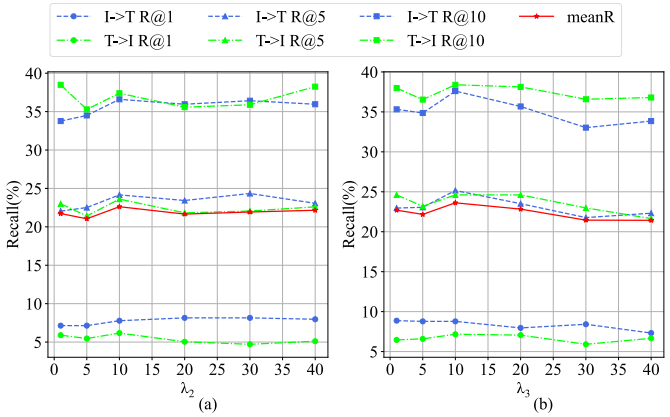


Fig. 6. Comparison of hyperparameters for loss function weights. The experiments are conducted on the RSICD dataset. (a) Step One. (b) Step Two.

TABLE VIII
COMPARISON OF DIFFERENT TEXT ENCODERS ON THE RSICD TESTSET

Backbone	Sentence Retrieval			Image Retrieval			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
CLIP-Text [63]	8.33	23.33	36.23	7.04	23.82	38.32	22.85
Skip-Thoughts [50]	9.24	22.51	36.23	6.70	24.13	39.14	23.26

observe the effects on the RSICD dataset when varying the weight of the InfoNCE loss $\lambda_2 \in \{1, 5, 10, 20, 30, 40\}$. The results indicate that the best performance is achieved when λ_2 is set to 10.

Figure 6(b) demonstrates the effects on retrieval accuracy on the RSICD dataset when varying the weight of the BCE loss $\lambda_3 \in \{1, 5, 10, 20, 30, 40\}$, while keeping the weights of step one fixed at $\lambda_1 = 1$ and $\lambda_2 = 10$. The findings suggest that an increase in λ_3 beyond ten leads to a decline in retrieval accuracy. This indicates that excessive emphasis on aligning the concept space also results in a shift in the model's focus. As a result, for our weight hyperparameter selection in step two, we set λ_1 to 1, λ_2 to 10, and λ_3 to 10 in the experiments.

7) Comparison of Different Text Encoders: As shown in Table VIII, we delve into the varying impacts of different text encoders on retrieval performance, involving the frozen contrastive language-image pre-training (CLIP) text encoder of ResNet-50 [63] and the skip-thoughts [50]. The results indicate that despite the abundance of pretraining data of CLIP, its performance on RSITR falls short of that achieved by skip-thoughts. The differences might be attributed to the skip-thoughts we utilized containing 1226 words tailored specifically for the RSICD dataset, whereas CLIP encompasses 49 152 words. Therefore, while CLIP's text encoder may find broader applicability across various domains, it may be less optimized for specific tasks. Consequently, we opt for skip-thoughts as the text encoder to ensure equitable comparisons with other methods and to optimize the cost of the text.

8) Analysis of the Training Strategy: To verify the effectiveness of our proposed training strategy, we conduct a comparison between the "one-step" training and the "two-step" training, where the former employs all the training objectives at the beginning. As shown in Table IX, the results indicate that our proposed two-step strategy significantly enhances the model's retrieval performance. In addition, during the training process, we discovered that employing BCE loss from the beginning would significantly slow down the convergence rate of the model. We propose that in the initial step, when semantic alignment is still incomplete, directly applying knowledge alignment may not be helpful in a chaotic semantic space. The concept space serves the purpose of aligning the values of the semantic space with those of multilabel classification. Thus, excessively or directly adjusting the weight of BCE loss would lead the entire model to overly focus on label information, neglecting the effectiveness of the semantic alignment.

TABLE IX
COMPARISON OF TWO TRAINING STRATEGIES ON THE RSICD TESTSET

Strategy	Sentence Retrieval			Image Retrieval			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
one-step	5.40	15.83	27.90	4.17	17.77	32.22	17.21
two-step	**9.24**	**22.51**	**36.23**	**6.70**	**24.13**	**39.14**	**23.26**

TABLE X
ANALYSIS ON INFERENCE EFFICIENCY

Approach	Params(M)	RSICD		RSITMD		NWPU	
		mR	IT(s)	mR	IT(s)	mR	IT(s)
VSE++ [7]	15	10.43	8.63	24.83	5.52	13.46	22.68
AMFMN [17]	36	16.42	25.56	29.72	6.39	32.22	148.41
GaLR [4]	46	18.96	22.92	31.41	6.23	-	-
KAMCL(Ours)	40	23.26	11.86	36.19	5.63	40.75	28.53

E. Analysis on Inference Efficiency

We further provide a detailed comparison of the inference efficiency of existing mainstream algorithms, including VSE++ [7], AMFMN [17], and GaLR [4], as shown in Table X. Compared with VSE++, although our approach entails a higher parameter count and lower inference efficiency, our performance has improved by 12.83%, 11.36%, and 27.29% on the three datasets, respectively. In comparison with the latter two methods, with similar parameter counts, our approach leads to an increase in inference efficiency by 48%, 10%, and 81%, as well as an improvement in retrieval performance by 4.30%, 4.78%, and 8.53%, respectively. Consequently, our model strikes an optimal balance between retrieval performance and inference speed.

F. Visualization and Analysis

In terms of sentence retrieval results as shown in Fig. 7, our model demonstrates superior retrieval results over other methods. Specifically, to more intuitively reflect the problems solved by our model, we include comparisons with the GaLR and AMFMN models. As indicated in the first row, our model pays more attention to the multilabel key information related to the shape of the number 8 in the image. While in the text, it is more inclined to find a sentence containing the description of the number 8. Though the third result with the number 8 is

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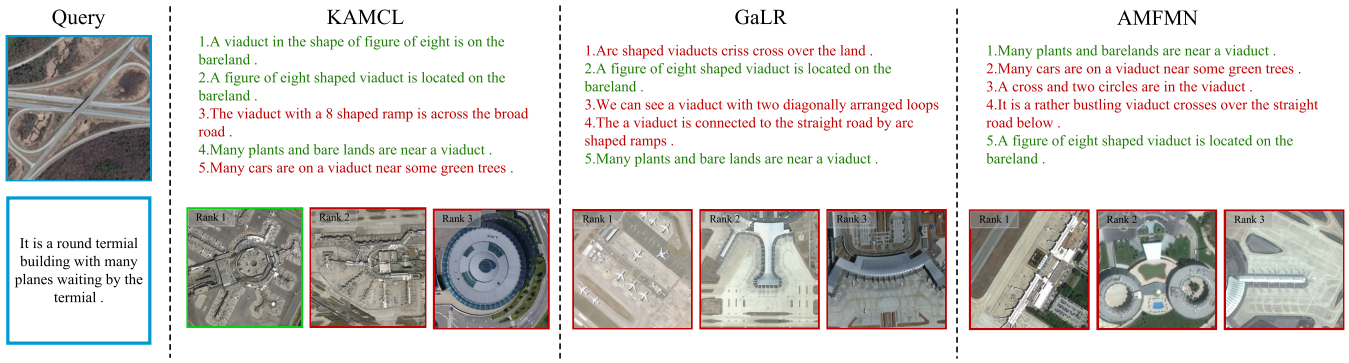


Fig. 7. Visualization of the qualitative sentence and image retrieval results of our KAMCL, GaLR [4], and AMFMN [17] on the RSICD dataset. For each sentence or image query, the matching results are marked as green, while the wrong ones are marked as red.

incorrect, our model is more likely to capture the information implied in the image.

In terms of image retrieval results, our KAMCL successfully retrieves the target images in the top-ranking position, while the other methods fail to retrieve them. In the example provided in the second row in Fig. 7, our model captures the key distinguishing concept "round" in the sentence, resulting in retrieved images that tend to contain information about round shapes. Though the Rank-3 retrieval results from KAMCL do not effectively highlight attributes like "plane" and "terminal," this oversight leads to the situation that the primary objective of searching for a plane terminal is overlooked, rather than concentrating on a round building. This discrepancy arises from our model's strong inclination toward the "round" attribute. In the context of retrieval tasks, it is noteworthy that words possessing distinctive attributes, like "round," frequently become the central focus of the query. Therefore, such retrieval logic is reasonable.

V. Conclusion

In this article, we have proposed a KAMCL method for RSITR, which includes a novel KAL framework for distinguishing analogous descriptions and embraces the MCL for enlarging the scale of negative samples. Particularly, the KAL framework includes knowledge initialization, construction, filtration, and alignment operations to achieve a better focus on the key concepts. Moreover, we designed an HA module to better capture the multilevel information for visual modality. Our method addresses the challenge of analogous descriptions in the remote-sensing domain and demonstrates improvements of 2.65%, 2.08%, and 8.53% on mR on the prevailing RSICD, RSITMD, and NWPU-Captions datasets with ResNet-18.

In the future, we plan to expand our method in two aspects. On the one hand, we intend to leverage external tools or prevailing vision-language pretraining models to enhance the representations of remote-sensing image–text pairs. On the other hand, we plan to take advantage of the image captioning capability of large multimodal models to reannotate remote-sensing datasets, thus obtaining more granular textual labeling information.

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